

CSE 4615:: Wireless Networks

Understanding Medium Access Control Mechanisms For Managing Shared Wireless Channels

Lecture 5

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Today

1. How does channel access differ between wireless and wired networks?

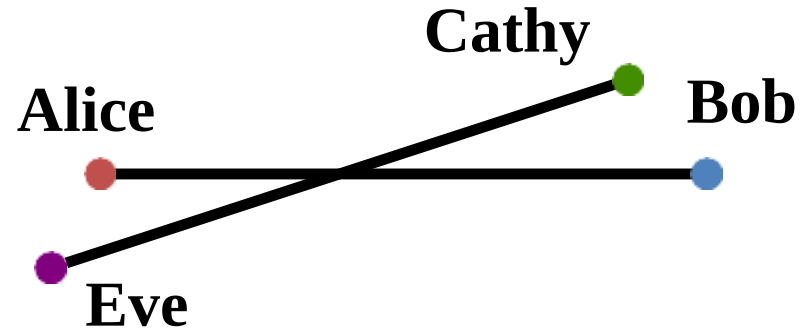
1. Medium access control: Provisioned protocols

1. Contention-based medium access control

Wireless is a shared medium

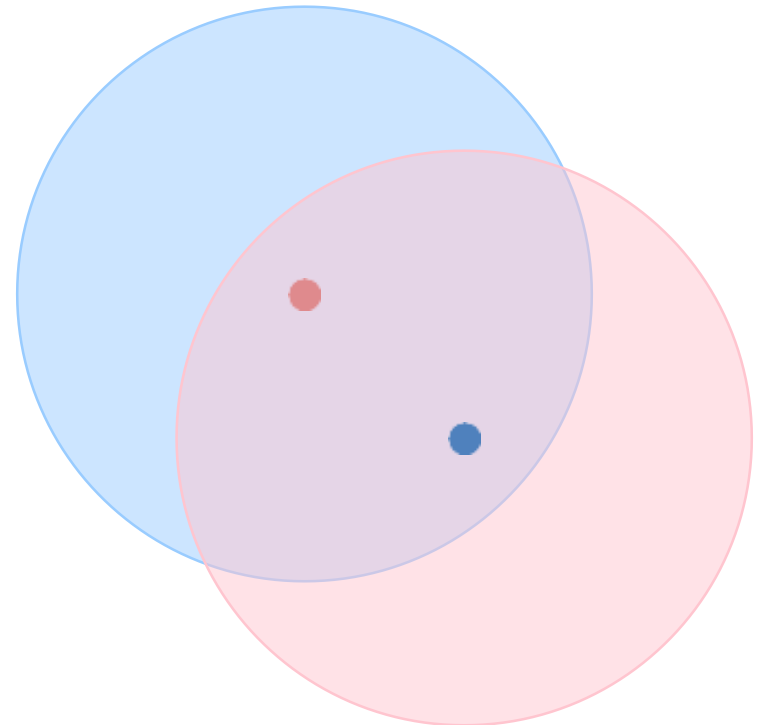
- **Wired networks:**

Alice and Bob's conversation is **independent of** Cathy and Eve's conversation



- **Wireless networks:**

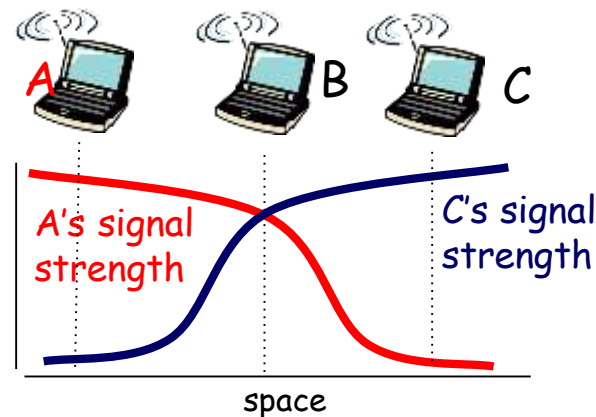
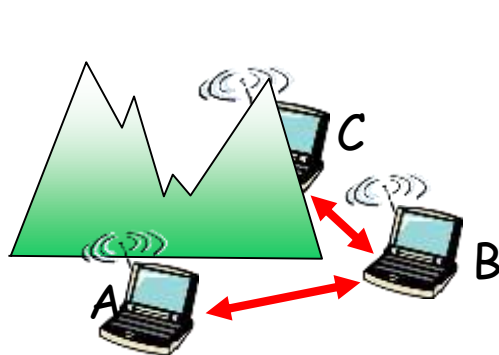
Close by wireless conversations **share the same wireless medium**



CSMA: Carrier Sense, Multiple Access



- Multiple access: channel is shared medium
 - Station: wireless host or access point
 - Multiple stations may want to transmit at same time
- Carrier sense: sense channel before sending
 - Station doesn't send when channel is busy
 - To prevent collisions with ongoing transfers
 - But, detecting ongoing transfers isn't always possible



Collision Avoidance & Collision Detection



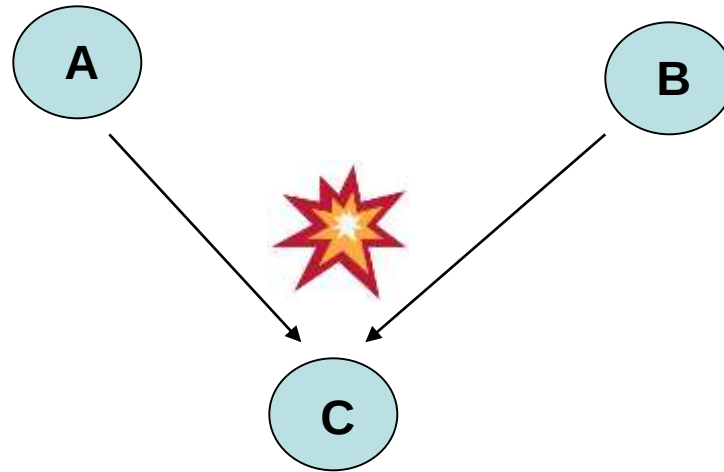
- Collision detection in wired Ethernet
 - Station listens (receives) while transmitting
 - Detects collision with other transmission
 - Aborts transmission and tries sending again
- Problem #1 in Wireless: cannot detect all collisions
 - Hidden terminal problem
 - Fading
- Problem #2 in Wireless: listening while sending
 - Strength of received signal is much smaller
 - Expensive to build hardware that detects collisions
- So, 802.11 does *not* do collision detection

Channel Access through CA: Collision Avoidance, Not Detection

- Listen to medium and wait until it is free (no one else is talking)
- Wait a random backoff time
- **Advantage:** Simple to implement
- **Disadvantage:** Cannot recover from a collision

Wireless Interference

- Two transmitting stations interfere with each other at the receiver
- Receiver gets garbage



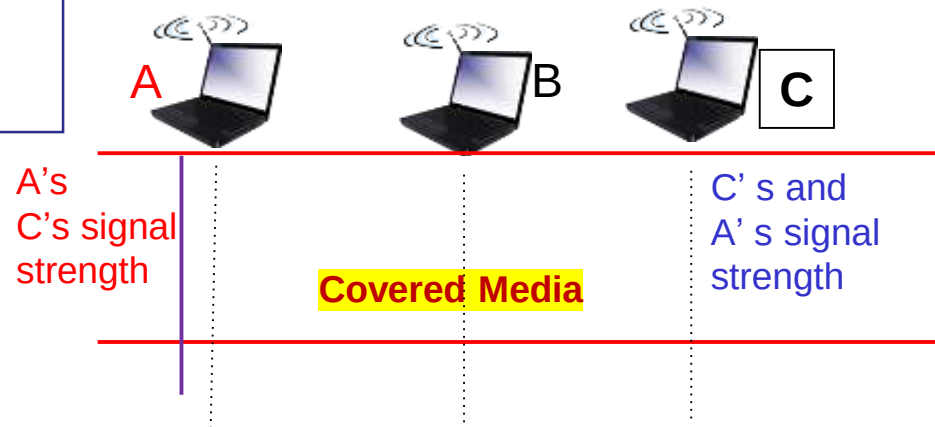
Collision Detection and CSMA-CD

- **Procedure**
 - Listen to medium and wait until it is free
 - Start talking, but listen to see if someone else starts talking too
 - If collision, stop; start talking after a random backoff time
- Used for hub-based Ethernet
- **Advantage:** More efficient than basic CSMA
- **Disadvantage:** Requires ability to detect collisions by transmitting station.
 - Require additional hardware antenna for concurrent receiving while transmitting (Radios are full duplex)
 - More difficult in wireless scenario due to significant signal attenuation

Collision Detection in Wired Network

Here, **Signal State** at Receiver (node B) and Transmitter (Node A) are **almost Identical**.
‘Fate sharing’ of the link’

Collision experienced at Receiver (node B) while both node A and C are transmitting



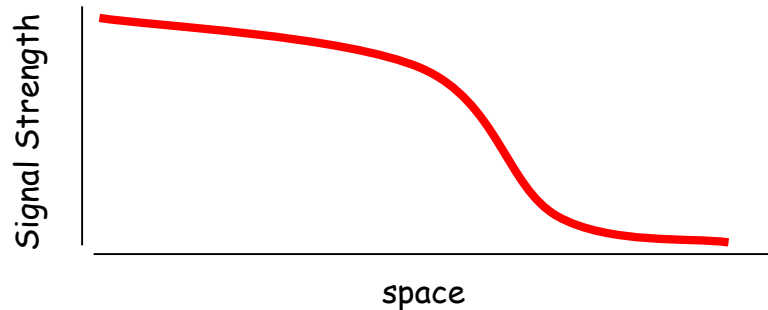
Hence, **Collision Detection at Transmitter (Node A) is Feasible**, by **sensing**(receiving) the **signal strength available now** by **additional receiving antenna** while transmission is going on.

Wireless Link Characteristics

Wireless is a broadcast medium!

Differences from wired link

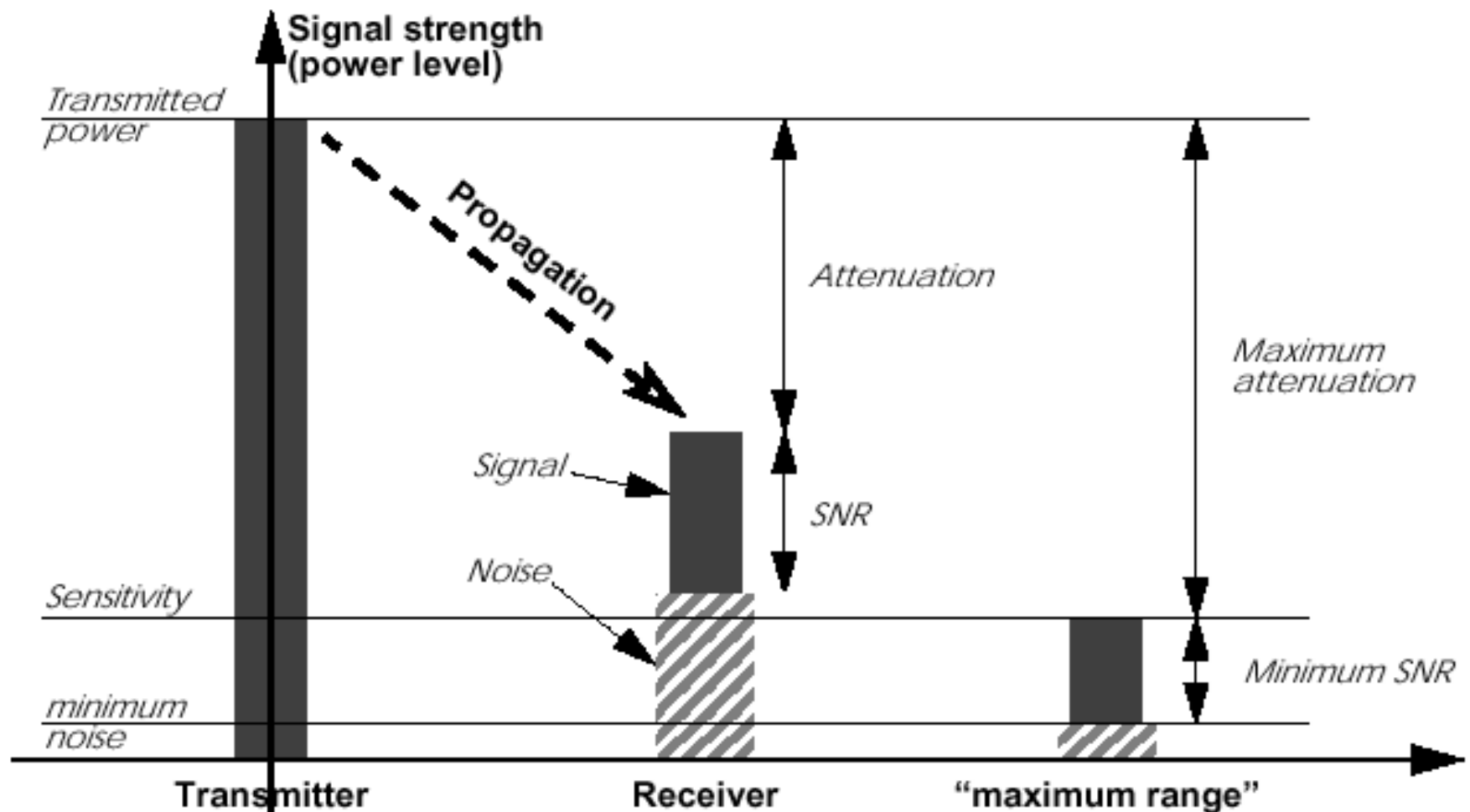
- **decreased signal strength:** radio signal attenuates as it propagates through matter (path loss)



- **interference from other sources:** standardized wireless network frequencies (e.g., 2.4 GHz) shared by other devices (e.g., phone); devices (motors, microwaves) interfere as well

.... make communication across (even a point to point) wireless link much more "difficult"

Wireless Link Characteristics: Attenuation & Propagation range

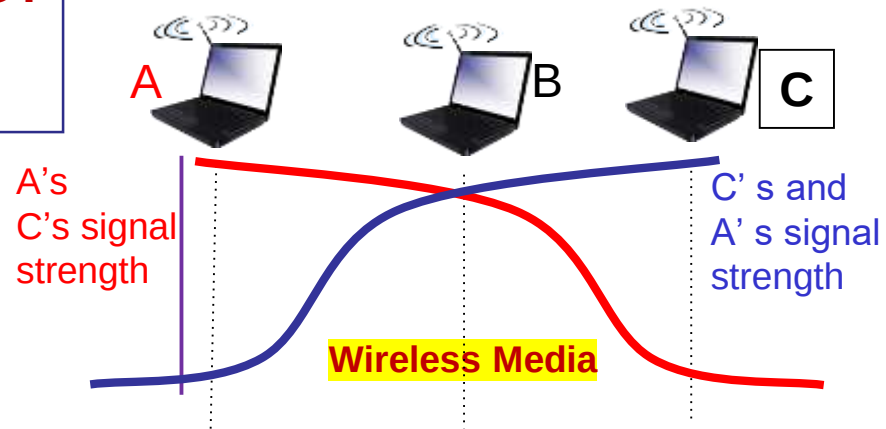


Collision Detection in Wireless Network

Here, **Signal State** at Receiver (node B) and Transmitter (Node A) are **NOT Identical**.

‘No Fate sharing’ of the link’

Collision experienced at Receiver (node B) while both node A and C are transmitting



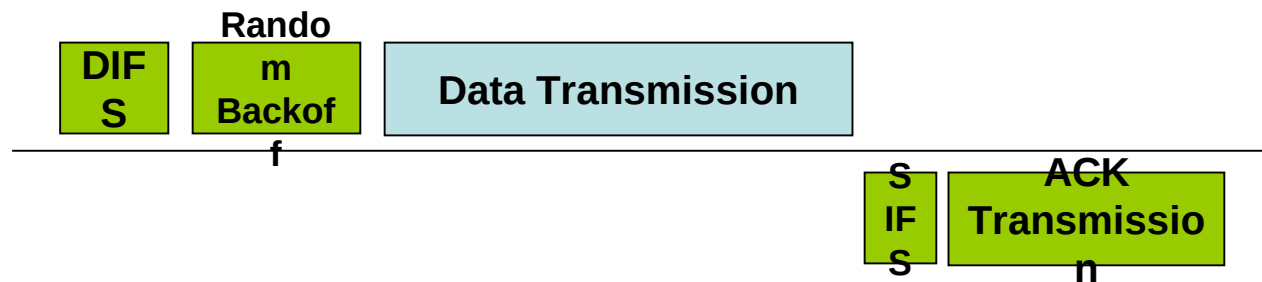
Hence, **Collision Detection at Transmitter (Node A) is NOT Feasible**, by **sensing**(receiving) the **signal strength available now** by **additional receiving antenna while transmission is going on**.

Failure to Detect Collision in Wireless

- No “fate sharing” of the link
 - High loss rates
 - Variable channel conditions
- Transmit signal is stronger than received signal
- Hence, Cannot take advantage of detecting multiple transmission by simultaneously transmit and receive
- Hence, Radios are not full duplex

Solution: Link-Layer Acknowledgments

- Absence of ACK from receiver signals packet loss to sender
- Sender interprets packet loss as being caused by collision



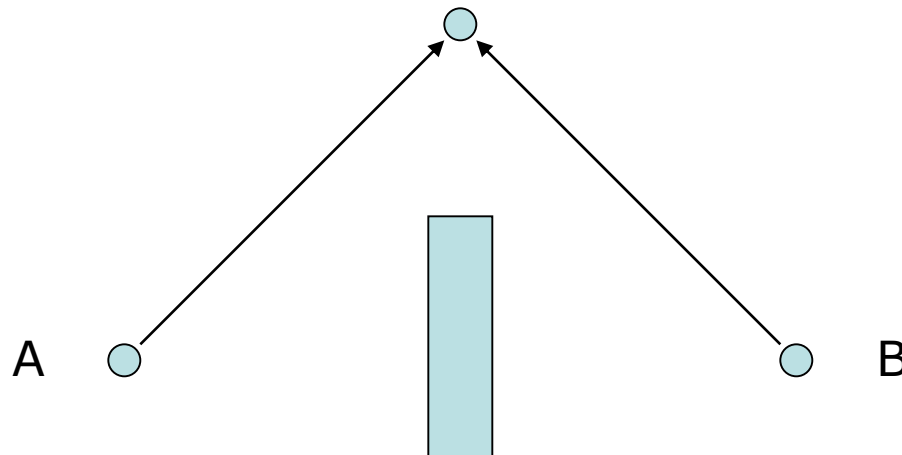
Problem: Does not handle hidden terminal cases.

Carrier Sense Multiple Access with Collision *Avoidance* (CSMA-CA)

- Similar to CSMA but *control* frames are exchanged instead of data packets
 - **RTS**: request to send
 - **CTS**: clear to send
 - **DATA**: actual packet
 - **ACK**: acknowledgement

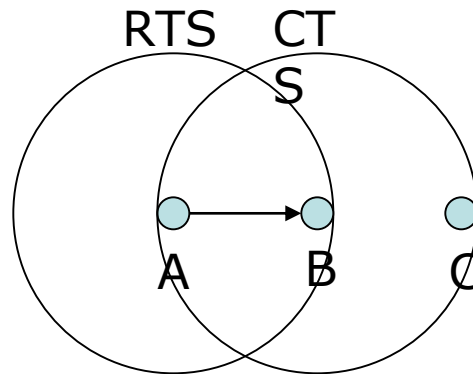
Carrier Sense Multiple Access with Collision Avoidance (CSMA-CA)

- Small control frames lessen the cost of collisions (when data is large)
- RTS + CTS provide “virtual carrier sense”
 - protects against hidden terminal



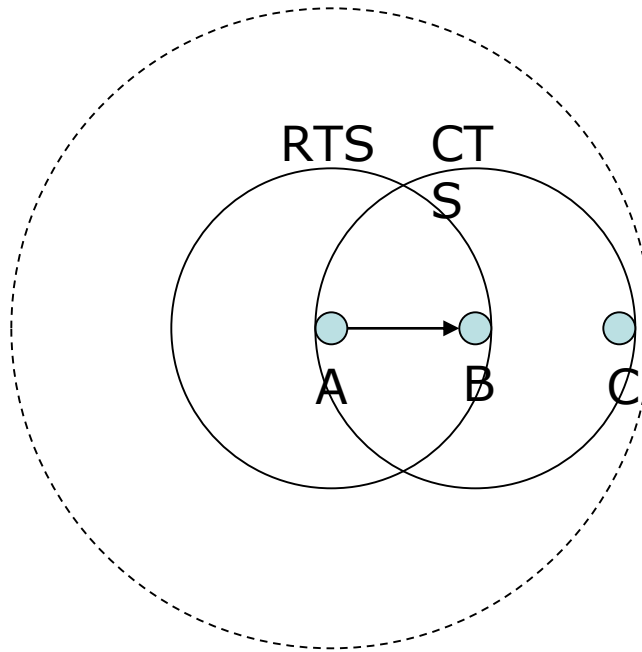
Virtual Carrier Sense

- Provided by RTS & CTS
- Prevents hidden terminal collisions



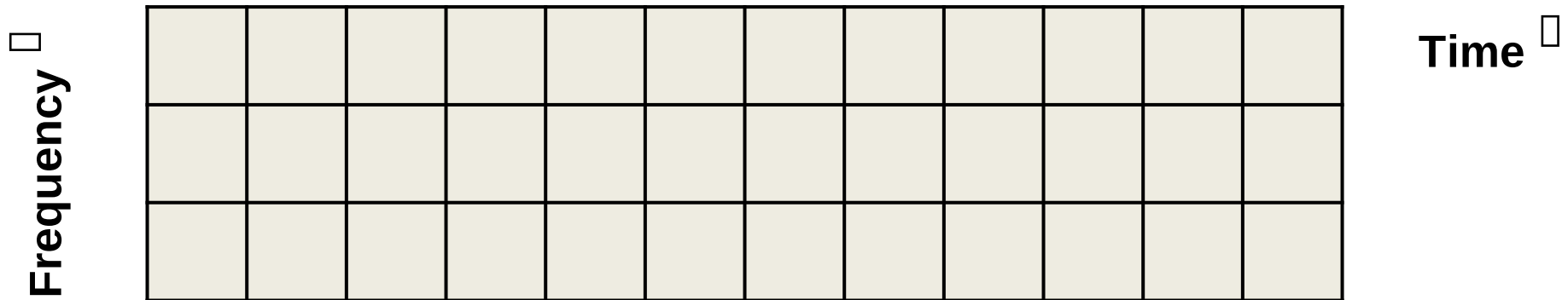
Hidden Terminal Revisited

- RTS transmission by **node A**, retrains all the **neighbors of A** from transmission during A->B transmission.
- RTS transmission by **node B**, retrains all the **neighbors of B** from transmission during A->B transmission.



Understanding Medium Access: The Problem

Medium access: The Problem



- Receiver must hear ≤ 1 strong transmission at a time, else **collision** (transmissions are lost)
- Two questions:
 1. **How** should the shared medium be **divided**?
 2. **Who** gets to use each such unit of division, and **when**?
- A **medium access control (MAC) protocol** specifies the

Today

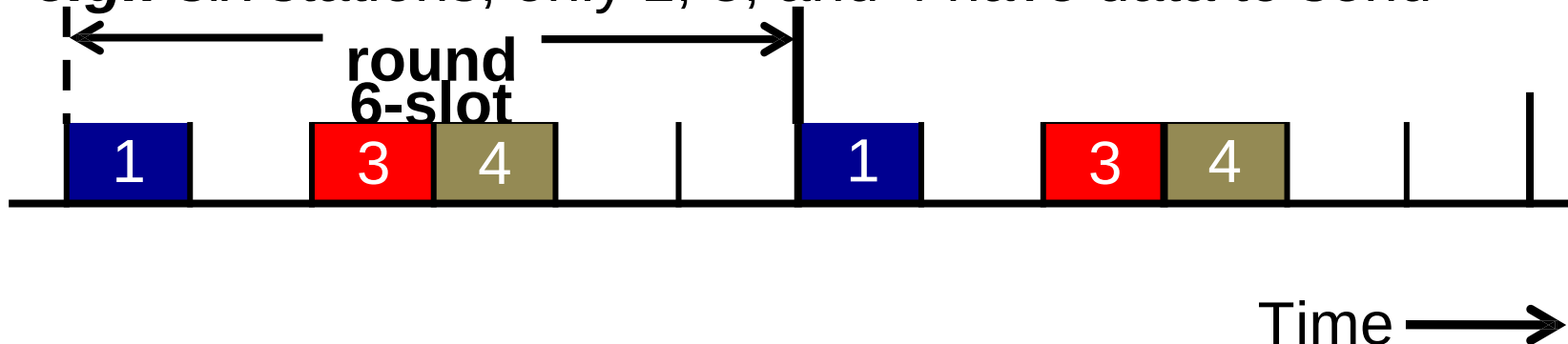
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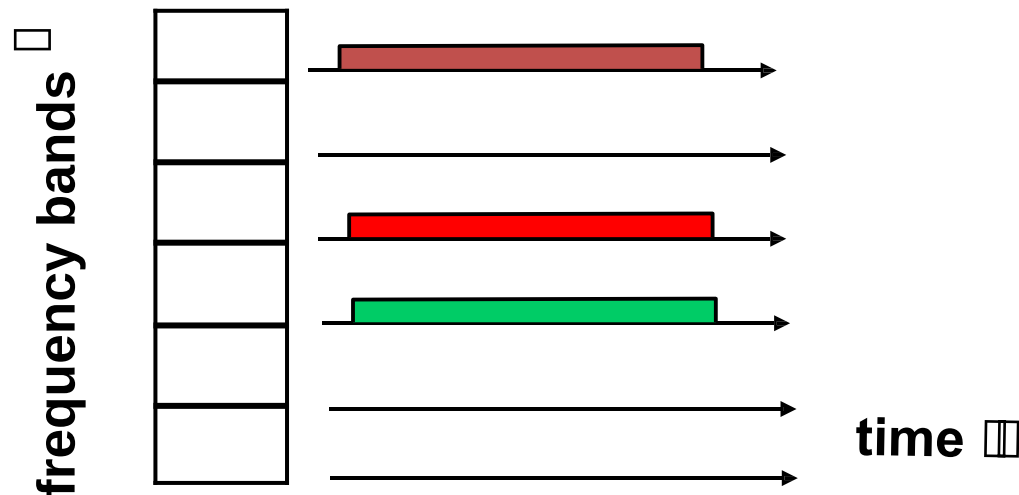
TDMA: Time Division Multiple Access

- Channel time is divided into **fixed-period, repeating rounds**
- Each user gets a fixed-length **slot** (packet time) in each round
 - Send across all frequencies during this slot
 - **Unused slots are wasted**
- **Out-of-band:** Mechanism for allocating/de-allocating slots
- **e.g.:** six stations, only 1, 3, and 4 have data to send



FDMA: Frequency Division Multiple Access

- Channel's frequency range (for all time) divided into frequency ***bands***
 - Each user gets a fixed frequency band (**unused frequency slots are wasted**)
- e.g.***: six stations, only #1, #3, and #4 have data to send



TDMA and FDMA: Considerations

- **Advantages**

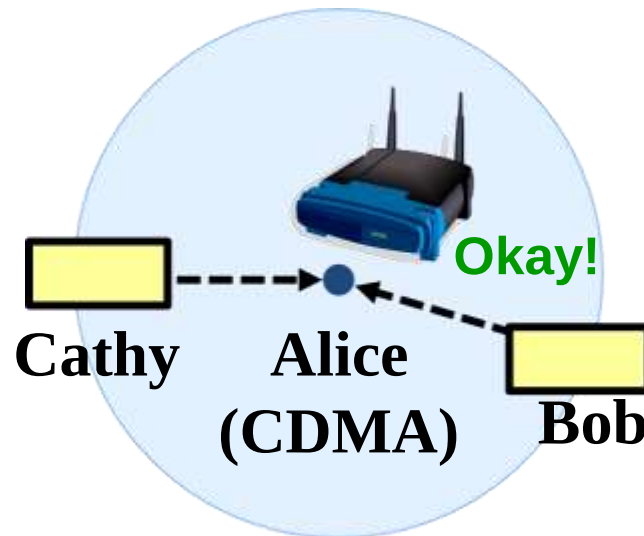
1. Users are **guaranteed** to be able to send bits, **continuously (FDMA)** or **periodically (TDMA)**

- **Disadvantages**

1. Unused time slots or frequency bands **reduce channel utilization**
1. An out-of-band mechanism is needed to allocate slots or bands (which **requires another channel**). Slot assignment in TDMA isn't automatic—it requires coordination over a separate signaling channel.
1. Guard bands or guard times **reduce channel utilization**

CDMA: Code Division Multiple Access

- All users transmit over the **same frequencies**, and **at the same time**:



- Allows multiple users to coexist and transmit simultaneously with **no interference**

Today

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Idea: Use a single, shared channel

- Spread transmission across whole 100 MHz of spectrum
 - **Remove constraints** assoc. w/one channel per user
 - No need for strict **one-channel-per-user** allocations.
 - Users can dynamically access any part of the shared channel, making it more efficient and flexible
 - **Robust to multi-path fading**
 - In wireless environments, signals reflect and arrive along multiple paths
 - Spreading the signal across many frequencies ensures that even if some parts are lost, others may arrive intact
 - **Supports peer-to-peer communication**
- **Collisions:** Receiver must hear ≤ 1 strong transmission at a time
- So adopt **deference** from Ethernet
 - **Listen** (*carrier sense*, **CS**) before sending, **defer** to ongoing

Concurrency versus Taking Turns

- Far-apart links should **send concurrently** (*spatial reuse*)

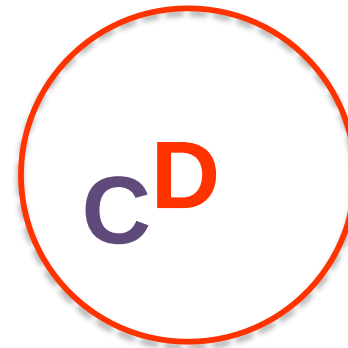
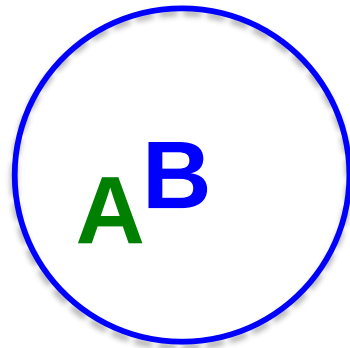


- Nearby links should **take turns**:



When Does CS Work Well?

- Two transmission pairs are **far away** from each other
 - **Neither sender** carrier-senses the other

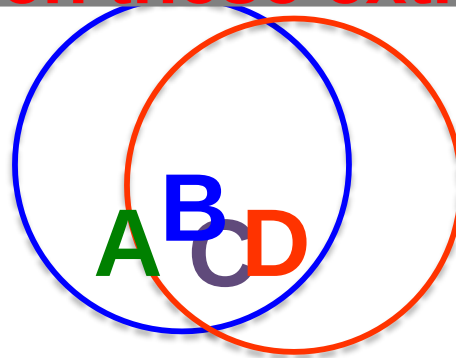


B transmits to A, **while** D transmits to C.

When Does CS Work Well?

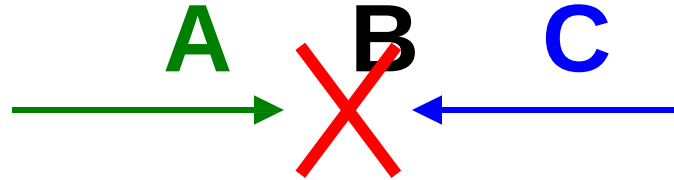
- Both transmitters **can carrier sense** each other
 - Carrier sense uses **thresholded correlation** value to determine if medium occupied

But what about cases in between these extremes?



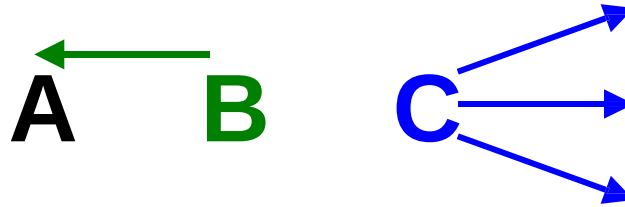
B transmits to A, D transmits to C, **taking turns.**

Hidden Terminal Problem



- **C** can't hear **A**, so **C** will transmit while **A** transmits – **Result: Collision at B**
- Carrier Sense insufficient to detect all transmissions on wireless networks!
- **Key insight:** Collisions are **spatially located at receiver**

Exposed Terminal Problem



- If C transmits, does it cause a collision at A?

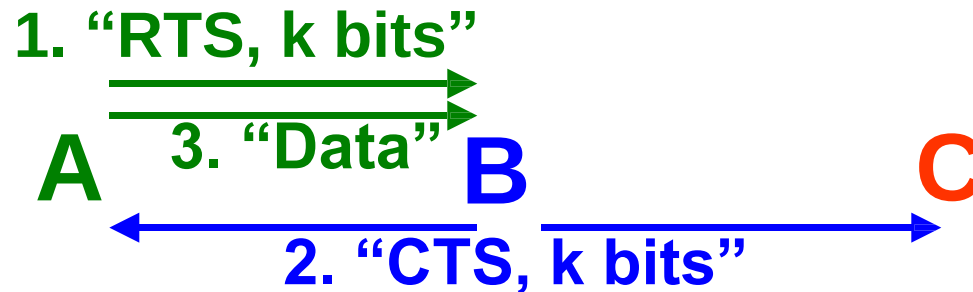
No!

– Yet C cannot transmit while B transmits to A

- Same insight: **Collisions spatially located at receiver**

RTS/CTS

- Exchange of two short messages: *Request to Send (RTS)* and *Clear to Send (CTS)*
- Algorithm
 1. A sends an **RTS** (tells B to prepare) with message length k
 2. B replies an **CTS** (echoes message length)
 3. A sends its **Data**



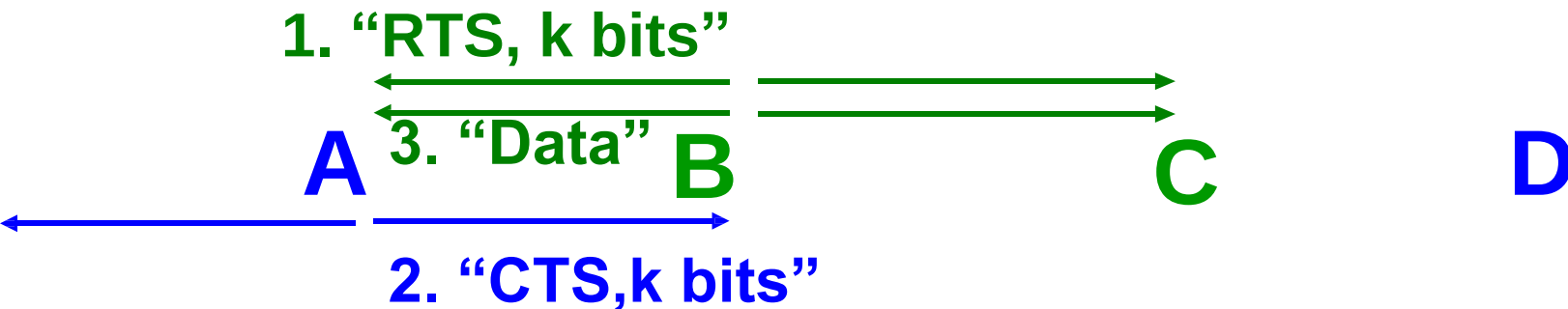
Deference to CTS

- **Hear CTS** □ Defer for **length of expected data** transmission time
 - **Solves hidden terminal problem**



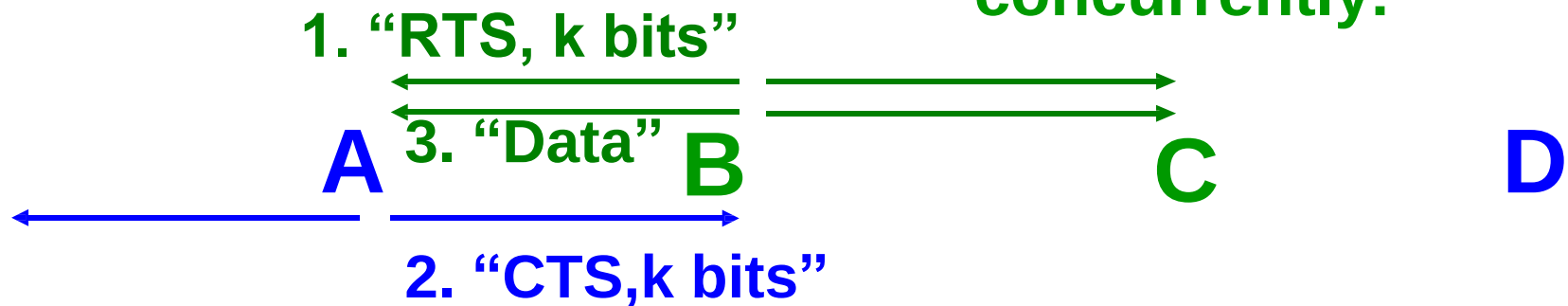
Deference to RTS, but not CS

- Hear -> **RTS** , -> **No CTS** , -> **Data**
 - So terminals **C** **Do not transmit to D !**
 - **But Could transmit to D !**



Deference to RTS, but not CS

So **exposed** terminals **B, C** can transmit concurrently:



✓ Common Solutions:

1. RTS/CTS Mechanism (Request to Send / Clear to Send)

- Node B sends an RTS to A.
- A replies with a CTS.
- Node C hears **RTS** from B but **does not hear CTS** from A (because A is out of C's range).
- ➤ **C knows it won't interfere with A** and can proceed with its own transmission safely.

✓ **Insight:** RTS/CTS provides **receiver-based collision control**, not just sender-based carrier sensing.